

Assignment 4 (3)

Monday 17 May 2021

Surname:

Initials:

Student number:

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

You are not allowed to use a calculator. Show all your steps when you do the following:

QUESTION 1 [3]

For what value of c is the function f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} cx^2 & \text{if } x \leq 2 \\ 3 + c & \text{if } x > 2. \end{cases}$$

QUESTION 2 [3]

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \frac{\cos(x^2)}{x^2}$$

QUESTION 3 [2]

Find $f(5)$ and $f'(5)$, if the tangent line to $y = f(x)$ at $(5, 4)$ passes through the point $(0, 2)$.

QUESTION 4 [4]

Find the derivative of the following function using the definition of a derivative.

$$f(x) = \frac{1}{ax + b}, \quad a, b > 0.$$

State the domain of the function and the domain of its derivative.